

High-Throughput Molecular Simulation of Commercial Li-Ion Electrolytes and Their Impact on Cell-Level Performance

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Abstract

1. Introduction

Electric energy storage systems, and lithium-ion batteries (LIBs) in particular, are central to the transition from fossil fuels toward renewable and electrified energy technologies. LIBs have become the dominant technology for electric mobility and are increasingly important for stationary energy storage, where they can help accommodate the intermittency and grid imbalances associated with renewable power generation. Nevertheless, further improvements in energy density, power capability, lifetime, safety, and charging time remain essential to broaden their application range and accelerate their market penetration. Among these challenges, fast charging is particularly demanding because increasing the applied current intensifies transport limitations throughout the cell and can substantially accelerate degradation.

Electrolyte transport plays a central role in these limitations. During battery operation, the electrochemical reactions at the porous electrodes generate salt-concentration gradients in the liquid electrolyte, which become increasingly pronounced at high charge/discharge rates and low temperatures. The resulting concentration polarization contributes to the overall cell overpotential and can lead to severe local electrolyte depletion. During fast charging, these transport limitations can promote a decrease of the negative-electrode potential toward conditions at which lithium plating becomes thermodynamically favorable, thereby compromising cycle life and safety. Consequently, an accurate description of electrolyte transport is required not only for predicting rate capability, but also for understanding and mitigating degradation mechanisms associated with demanding operating conditions.

Ionic transport through a battery electrode is governed by both the porous microstructure and the intrinsic transport properties of the electrolyte. While electrode porosity and tortuosity determine the effective transport pathways available to the ions, the response of the liquid electrolyte itself depends on its conductivity, salt diffusion coefficient, cation transference number, and thermodynamic factor. Within the concentrated-solution framework underlying the widely used Newman porous-electrode model, these properties collectively determine the development of electrolyte concentration and potential gradients. Importantly, they depend strongly and nonlinearly on electrolyte composition, salt concentration, and temperature. Reliable cell-level predictions therefore require these

quantities to be accurately parameterized over the range of thermodynamic conditions experienced during battery operation.

This requirement becomes particularly important when electrolyte formulation is considered as part of battery optimization. Improving electrolyte transport properties represents an important route toward faster charging and robust operation over a wider temperature window. However, electrolyte design involves a vast chemical and compositional space encompassing different salts, solvents, solvent ratios, additives, concentrations, and operating temperatures. At the same time, electrolyte optimization cannot be considered independently from electrode and cell design. Parameters such as electrode thickness, porosity, active-material loading, particle dimensions, and operating current interact with electrolyte transport to determine the resulting concentration polarization and cell performance. Physics-based cell models provide an attractive means of exploring this multidimensional design space and rapidly evaluating the consequences of changes in materials and electrode architecture. Their predictive capability, however, ultimately depends on the availability and quality of the underlying material parameters, with electrolyte transport properties representing one of the most demanding parameterization requirements.

Important experimental efforts have therefore been devoted to establishing consistent transport-property datasets for battery electrolytes. In particular, the systematic studies of Landesfeind and Gasteiger demonstrated the determination of ionic conductivity, binary salt diffusion coefficients, lithium-ion transference numbers, and thermodynamic factors as functions of concentration and temperature for technologically relevant carbonate-based electrolytes. Such datasets are directly compatible with the concentrated-solution theory used in Newman-type battery models and have consequently become valuable references for physics-based cell simulation. Their generation, however, requires the combination of several dedicated experimental techniques and extensive measurements across concentration and temperature. Repeating this procedure for every new electrolyte formulation therefore becomes prohibitively demanding when moving toward high-throughput exploration of electrolyte chemical space.

Molecular dynamics (MD) simulations offer an attractive complementary route because they can probe transport directly from the molecular motion of ions and solvents and can, in principle, be applied systematically across large sets of electrolyte compositions. However, the quantities most commonly extracted from classical MD simulations are the self-diffusion coefficients of the individual ionic species, obtained from their mean-square displacements, and ionic conductivities estimated through the Nernst–Einstein relation. These quantities are not, in general, equivalent to the transport coefficients required by concentrated-solution theory. The Nernst–Einstein approximation treats ionic motion as effectively uncorrelated, whereas battery electrolytes at technologically relevant concentrations can exhibit substantial cation–anion, cation–cation, and anion–anion dynamical correlations arising from ion association and collective transport. As a result, self-diffusion coefficients alone cannot rigorously determine the salt diffusion coefficient, the solvent-reference-frame transference number, or the ionic conductivity required by Newman-type models. A consistent connection between molecular dynamics and continuum electrolyte transport instead requires the collective Onsager transport coefficients, which can be transformed into the corresponding Stefan–Maxwell transport description, together with the appropriate thermodynamic contribution governing non-ideal solution behavior. [here give a few words about how classical MD reports

self-diffusion coefficients mostly and that those are not correct from the perspective of concentrated solutions]

In this work, we develop, automate and experimentally validate a molecular-dynamics-based framework for obtaining the complete set of electrolyte transport and thermodynamic parameters required by Newman concentrated-solution theory. Rather than relying on self-diffusion coefficients as surrogates for macroscopic ionic transport, the framework explicitly accounts for correlated species motion through an Onsager/Stefan–Maxwell description and provides conductivity, salt diffusion coefficient, lithium-ion transference number, and thermodynamic factor in a form directly transferable to physics-based battery models. By establishing a rigorous molecular-to-continuum connection and benchmarking the resulting parameters against experimental measurements, the proposed methodology provides a route toward high-throughput evaluation of electrolyte formulations while retaining the transport description required for predictive cell-level simulations. Ultimately, this framework enables electrolyte chemistry to be incorporated as an explicit design dimension in model-based battery optimization, connecting molecular-scale electrolyte formulation with cell-level performance, fast-charging capability, and operating-temperature limitations.

- General intro on Li-ion batteries
- Electrolyte transport controls rate capability, fast charging, low-temperature performance, ...
- Physics Based models, introduction, importance, etc.
- Problem of parameterization. High cost, problems associated with electrolyte characterization in commercial cells (problem with recovery).
- Experimental datasets are limited.
- MD as an alternative. Challenges for quantitative agreements with MD and experiments. Many MD studies report self-diffusion, which does not correspond to salt-diffusion coefficient in concentrated electrolytes which require more rigorous treatment based on Stefan-Maxwell concentrated solution theory which limits the direct use of the properties in P2D models.
- Contribution of this work, automated library for MD computation of lithium ion electrolytes, rigorous treatment of transport properties, direct inputs and performance comparison shown in P2D models. MD simulations validated with available experimental data.

2. Computational Methods

2.1 Automated workflow

- Input composition
- Electrolyte system generation
- Force field parameters
- Run MD
- Analyse trajectories.
- Extraction of transport properties (conductivity, onsager coefficients, activity, Born numbers, etc.)

2.2 Included electrolyte systems

- EC, DMC, EMC,

- Salts: LiPF₆,...
- Binary/Ternary solvent mixtures
- Systems used for validation and for exploration

2.3 Physics Based Models

- General information about differential equations
- Reference cell used for validation
- Implementation

3. Results

3.1 Transport properties

- Temperature and concentration dependent curves of transport properties
- Fitting to polynomial (concentration) and arrenius type (temperature) functions
- Validation with experimental data
- Discussion about reproducibility between different simulations
- Consideration behind selection of partial charges
- Discussion of the importance of considering quantifying CIP,SSIP, etc. for the computation of the dipole moments (related to Persson paper)

3.2 Effect on electrochemical response

- Effect of transport properties on electrochemical response
- Sensitivity
- Effect of activity coefficient (maybe it is the chance to dig deeper into this @Laura)
- C-rate capability, HPPC, EIS for a complete set of simulated electrolytes. (to decide on the selection)

4. Discussion

Author Contributions

Acknowledgements

Competing interests

Data availability

Code availability

Supplementary Information

References